



ANALYSTLAB AFRICA

Experience Lab Internship

Project Week 2 Task

FinTrust Digital Bank: Moving from Planning to Practical Execution

Data Analytics Track

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Project Overview

FinTrust Digital Banking is undertaking a strategic initiative to enhance its financial intelligence capacities and digital banking support infrastructure. As part of the project, Week 2 project will be building on Week 1, business understanding and initial planning. This week project represents a pivotal transition in the project life-cycle, by moving from strategic planning to practical execution.

Week 2 Objectives

This week's objective is to move from planning into execution by performing an initial analysis of the FinTrust reviewed datasets, being FinTrust Customer and Transaction datasets. The goal is to clean the data, answer key business questions using SQL, develop exploratory visualizations in Python, build an initial management dashboard in Power BI, and Document on the business findings discovered.

Had successfully cleaned and prepared all required FinTrust datasets, identifies and addressed data quality issues, developed and executed 8 SQL queries to analyze customer segments and transaction behavior, created 7 key visualizations and revealed Business insights about them in Python, and Built a foundational Power BI dashboard with 6 KPIs and 8 main visuals.

KeyData Findings

1. Analysis revealed that about 90% of transactions made by customers were successful, representing the highest rate of transaction success rate.
2. We also discovered that about 62% of customers prefer using Mobile App for most of their transactions, but this same Mobile App accounts for 5.80% of transaction failure rate, which represents the highest.
3. Analysis also revealed that Transfers brings into the company the highest money (NGN218M) as a product, while Airtime/Data accounts for the smallest (NGN9M). This shows the products that actually drives transaction values.

Challenges & Decisions

A key challenge was handling the *Transaction_DateTime* Column, when cleaned data was loaded into the Power BI. The whole column returns as error and it was realized that the date was disarrange as [mm/dd/yyyy], which makes it difficult for Power BI to identify that a date instead of [dd/mm/yyyy] so it was decided to change to the original date for Power BI to recognize as a real date field. This allowed the analysis to proceed without any negative consequence.

Week 1 Plan Review

Week 1 objective: Week 1 project focused on understanding FinTrust Digital Bank Business and transforming raw customer and transaction data into a strategic business and data intelligence plan, which serves as a master-plan for the Data Analytics Track for the next four weeks of project work.

Implemented Activities: The activities actually performed this week were Business understanding review, Data review, Strategic analytical questions to solve, KPIs planning, Dashboard planning, and initial plan for week 2 to week 4.

Data Analytics Track

Part A: Data Preparation & Quality Assessment

This part seeks to address the initial data-quality assessments.

1. Data Profiling Summary:

Dataset	Records	Columns
FinTrust_Customer_Data.csv	1,500 + 1 header record	12
FinTrust_Transaction_Data.csv	12,000 + 1 header record	11

2. Data Quality Assessment:

Issue Identified	Details	Action Taken	Dataset
Missing values	Both <i>Device_Type</i> and <i>Location</i> fields were having 96 transaction records missing in each field.	All empty cell are changed to filled with “Unknown”	FinTrust_Transaction_Data.csv
Duplicates	0 duplicate records found across both datasets	Data was already unique so no action needed to take.	FinTrust_Customer_Data.csv FinTrust_Transaction_Data.csv
Incorrect or inconsistent values	All categorical fields had valid, consistent labels	No action needed to take.	FinTrust_Customer_Data.csv FinTrust_Transaction_Data.csv
Data-Type Issues	All fields matched their expected type (date as dates, amounts as numbers, IDs as text). There’s no mismatched.	No action needed to take.	FinTrust_Customer_Data.csv FinTrust_Transaction_Data.csv
Usual Values/Outliers	1,474 transaction in Amount_NGN fell above the upper bound (NGN117,922.86) using the IQR method.	Retained, since these reflect genuine high-value transactions, not data errors.	FinTrust_Transaction_Data.csv
Dataset Relationship	Every customer had at least one transaction.	No action needed to take.	FinTrust_Customer_Data.csv FinTrust_Transaction_Data.csv

Deliverable: [FinTrust Data Quality Assessment.xls](#)

Part B: SQL Business Analysis

This part answers about 8 meaningful business questions using SQL, covering data integrity, transaction value, customer segments, and monthly activity trend.

The following are the key 5 business questions answered using SQL.

Question 1: Data relationship check: do the two datasets join cleanly?

Results:

```
customer_records  transaction_records  orphan_transactions  customers_without_transactions
1500              12000              0                    0
```

Business Interpretation: This analysis identifies every transaction matches a customer and every customer has at least one transaction (0 orphans, 0 inactive customers), so joins on Customer_ID are safe and no records are lost. Customers average exactly 8.0 transactions each.

Question 2: What share of transactions succeed, fail, reverse or stay pending, and how much value sits in each status?

Results:

```
Transaction_Status  txn_count  pct_of_txns  total_value_ngn
```

```
--      Successful      10856      90.47      510807441
--      Failed           630       5.25      24825751
--      Reversed         326       2.72      15626118
--      Pending          188       1.57      9218044
```

Business Interpretation: 90.5% of transactions succeed. Failed, reversed and pending transactions make up 9.5% of volume and about NGN 49.7M (roughly 8.9% of all value). Failed plus reversed alone is 8.0% of volume, which is the main workload for customer support. All value analysis below uses Successful transactions only, so that failed or reversed money is not counted as real activity.

Question 3: Which transaction types generate the most value, and how skewed are the amounts?

Results:

Transaction_Type	txn_count	total_value_ngn	avg_value_ngn	median_value_ngn
Transfer	3213	218365428	67963	15126
Deposit	1211	119531761	98705	21753
Card Purchase	2745	77957947	28400	7558
Cash Withdrawal	1290	61396432	47594	17784
Bill Payment	1336	24971374	18691	7026
Airtime/Data	1061	8584499	8091	2464

Business Interpretation: Transfers are FinTrust's core product: 42.7% of successful value (NGN 218M) from 3,213 transactions. Deposits have the highest average (NGN 98.7k) but a median of only NGN 21.8k, and every type has a mean roughly 2.7 to 4.5 times its median. A few very large transactions pull the averages up, so medians should be used when describing a typical transaction. Airtime/Data is high volume but only 1.7% of value.

Question 4: Do customer segments differ in transaction behaviour?

Results:

Customer_Segment	customers	txn_count	txns_per_customer	value_per_customer_ngn
SME	216	1553	7.2	354889
Student	278	2044	7.4	350292
Premium	295	2148	7.3	342538
Everyday	711	5111	7.2	331535

Business Interpretation: Not much. Value per customer ranges only from NGN 331.5k (Everyday) to NGN 354.9k (SME), and transactions per customer sit between 7.2 and 7.4. The current segments do not explain how customers behave. FinTrust may get more from a behaviour-based segmentation (recency, frequency, value) than from the existing labels.

Question 5: How does transaction activity change month to month?

Results:

month	txn_count	active_days	txns_per_day	avg_successful_ngn
2026-01	4133	31	133	46040
2026-02	3734	28	133	47191
2026-03	4133	31	133	47934

Business Interpretation: Monthly counts are 4,133, 3,734 and 4,133, but February has fewer days. Per day, volume is a constant 133 transactions in every month, and average successful value rises only slightly (NGN 46.0k to NGN 47.9k, +4.1%). There is no growth or seasonality to report over these 3 months. The timestamps are evenly spaced, which is a limitation of the synthetic data (see the Python notebook).

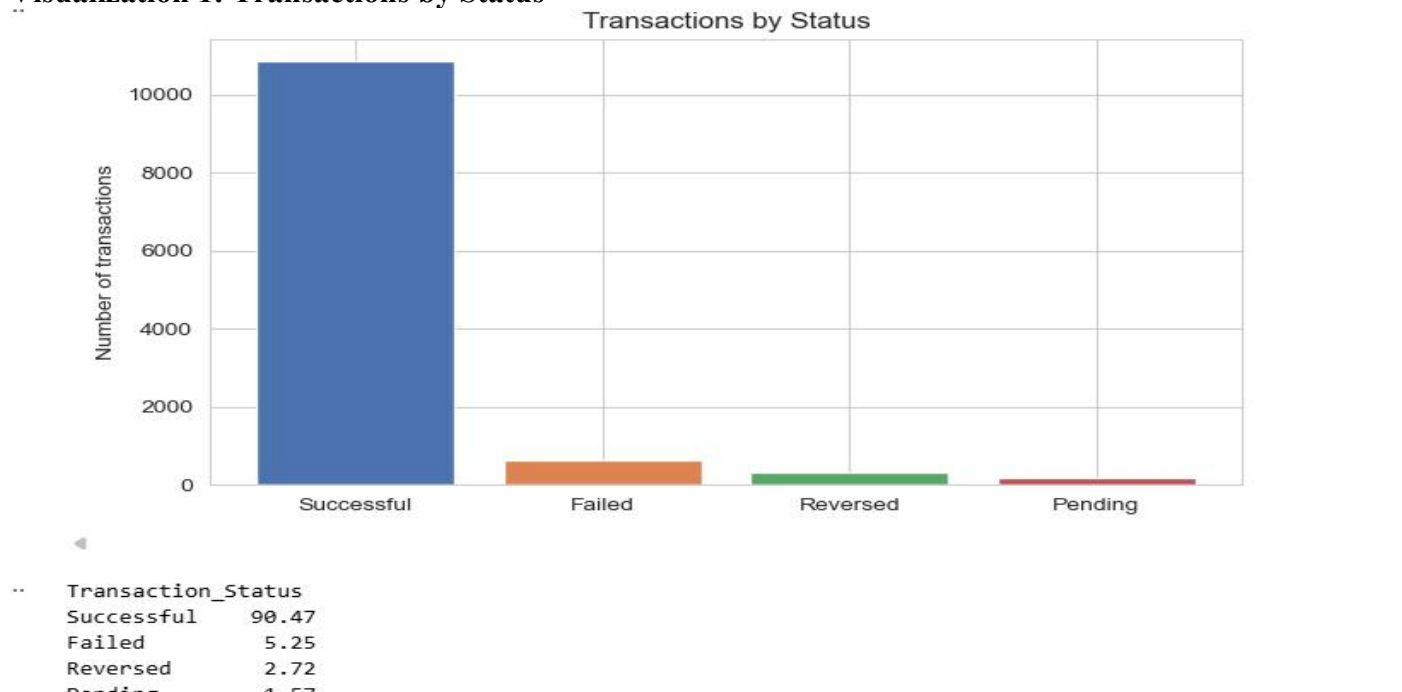
Deliverable: [*FinTrust Week2 SQL Analysis.sql*](#)

Part C: Python Exploratory Data Analysis

This part presents the exploratory analysis carried out on the cleaned Fintrust datasets using python.

Python (Pandas, Matplotlib, Seaborn) was used to conduct exploratory analysis and generate the following 7 key visualizations.

Visualization 1: Transactions by Status

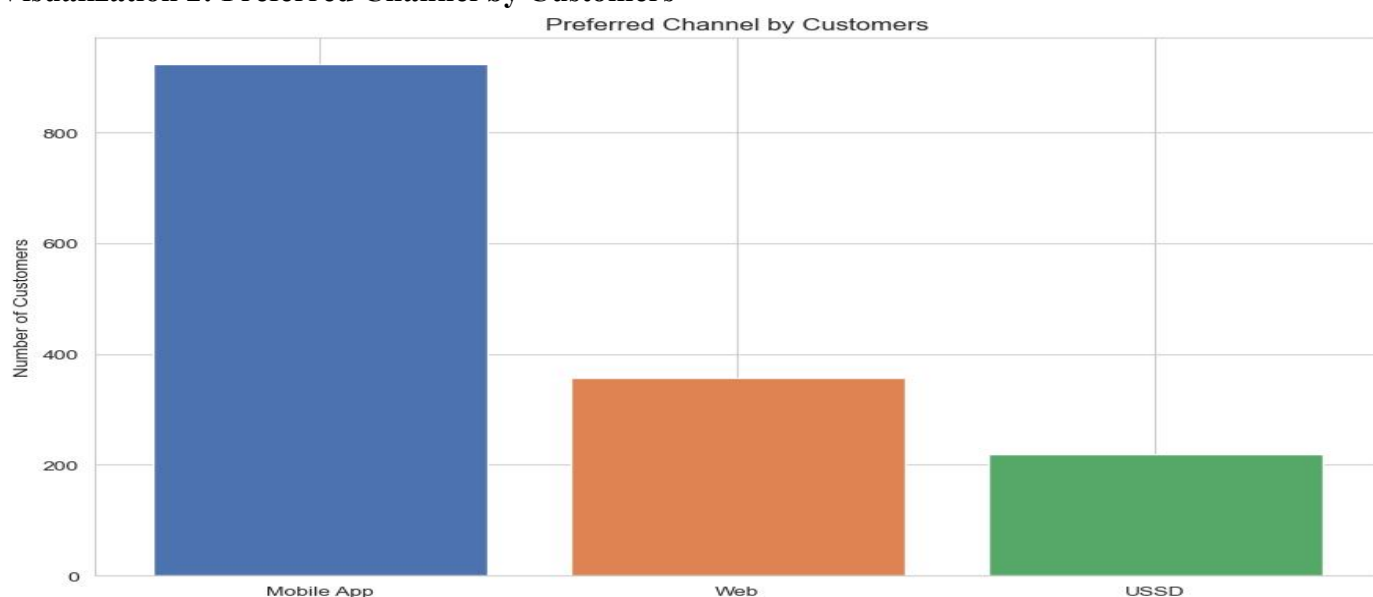


What it shows: 90.47% of transactions are successful, 5.25% Failed, 2.72% Reversed and 1.57% Pending

Why it matters: The success rate is the headline health measure of a digital bank, and every non-successful transaction is a potential customer_support contact.

Business Insight: The 9.5% of failed/reversed/pending transactions is the real workload for the support teams to reduce.

Visualization 2: Preferred Channel by Customers



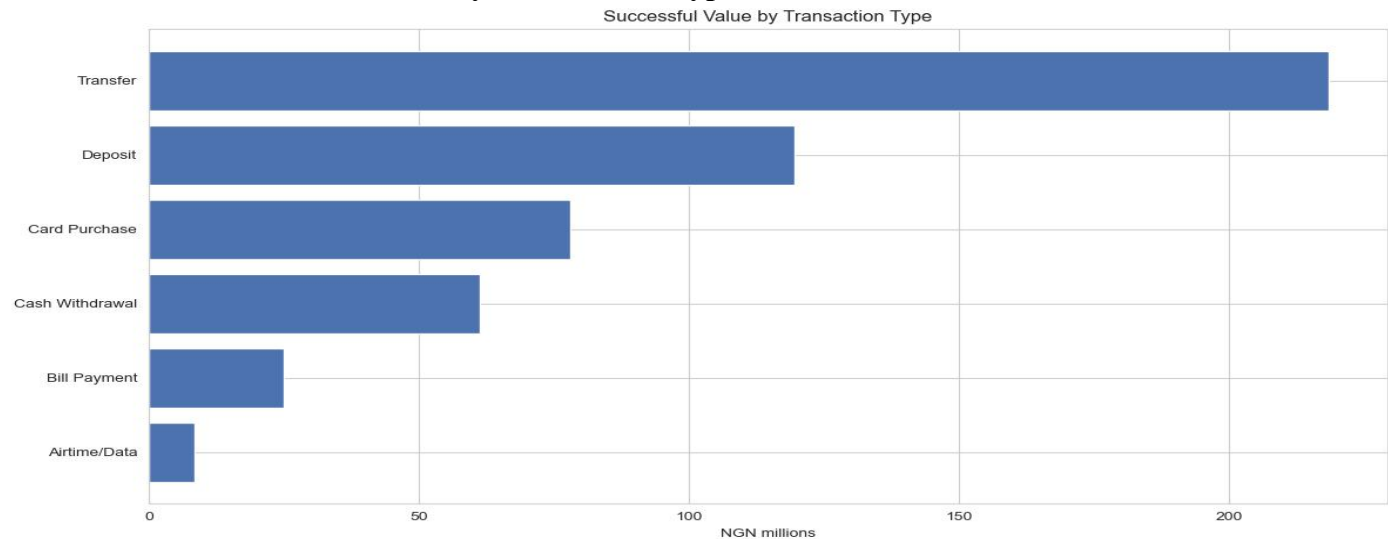
```
Preferred_Channel
Mobile App    61.6
Web           23.8
USSD          14.6
Name: count, dtype: float64
```

What it shows: The most number of customers(61.6%) prefer using Mobile App, followed by Web (23.8%) and USSD (14.6%).

Why it matters: It tells FinTrust where customers expected to be served.

Business Insight: FinTrust should consider investing a lot in Mobile App by making it more reliable and add more features, since its what most customers chose for most of their transactions.

Visualization 3: Successful value by Transaction Type



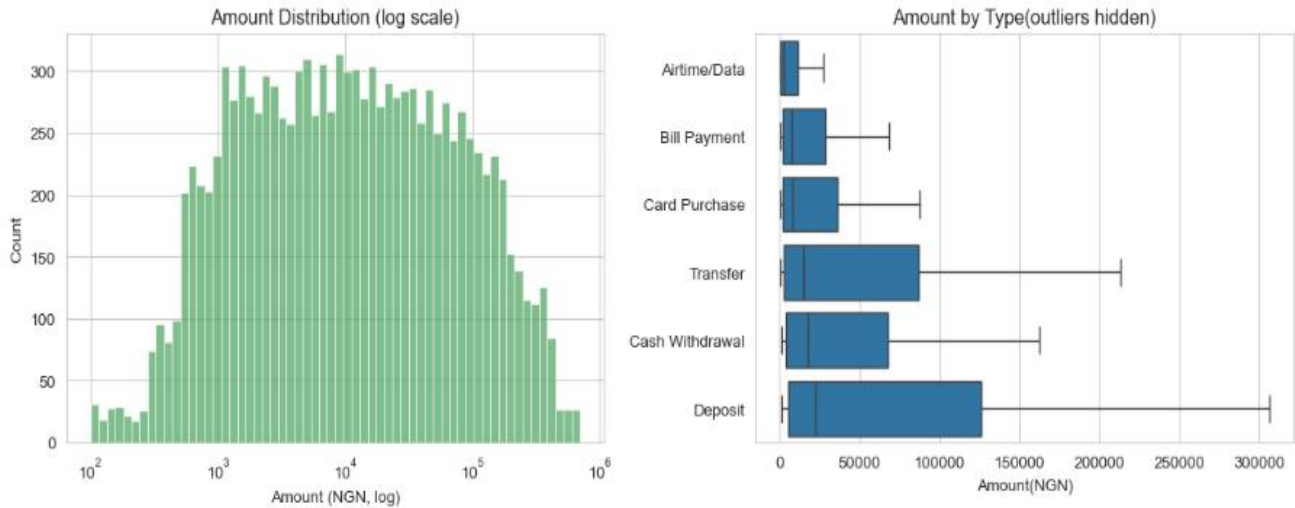
```
Transaction_Type
Airtime/Data      9.0
Bill Payment     25.0
Cash Withdrawal  61.0
Card Purchase     78.0
Deposit          120.0
Transfer         218.0
Name: Amount_NGN, dtype: float64
```

What it shows: Transfer generated the most money(NGN218M), followed by Deposit(NGN120M); Airtime/Data is the smallest (NGN9M)

Why it matters: It shows which products actually drives FinTrust's transaction value.

Business Insight: Looking at this, Transfers are the core product customers mostly subscribe to. FinTrust must keep this product more rliable and faster for customer so that it can brings in more value into the company.

Visualization 4: Distribution of Transaction Amount



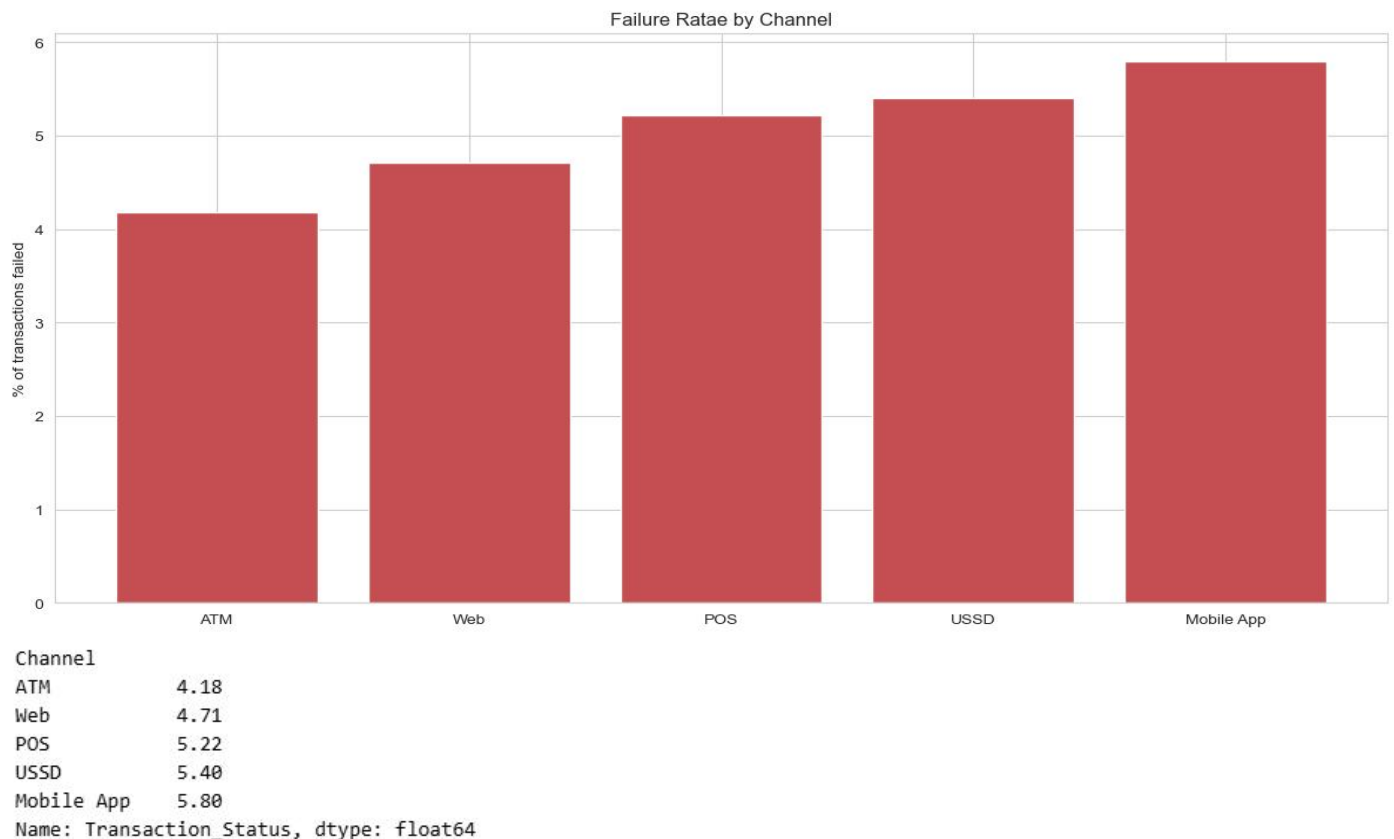
Mean 46,706 | Median 10,306 | Skewness 3.20
IQR upper bound 117,923 | Outliers 1,474 (12.3%) | Max 693,454

What it shows: Amounts are heavily right skewed (Skewness about 3.20%). Most transactions are small, with a long tail up to about NGN693K.

Why it matters: A few big numbers can make the "average" transaction look misleading.

Business Insight: Report the median (NGN10,306), not the mean, when describing a typical transaction.

Visualization 5: Distribution of Transaction Amount

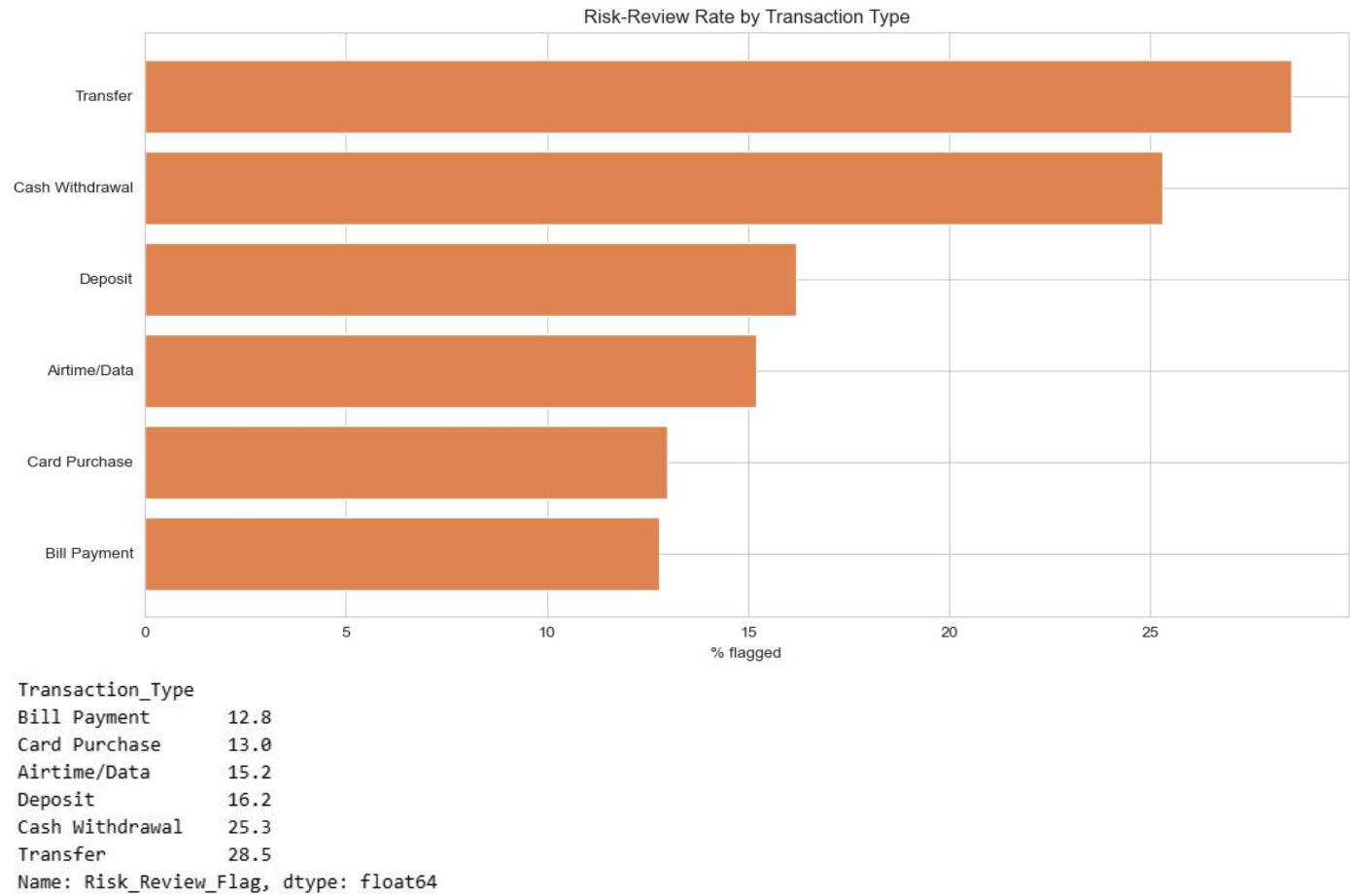


What it shows: Mobile App has the highest failure rate (5.80%); ATM has the lowest (4.18%).

Why it matters: Since Mobile App is the biggest channel used, its failures affect the most Customers.

Business Insight: Fixing Mobile App reliability would remove the most failed transactions overall.

Visualization 6: Risk-Review Rate by Transaction Type

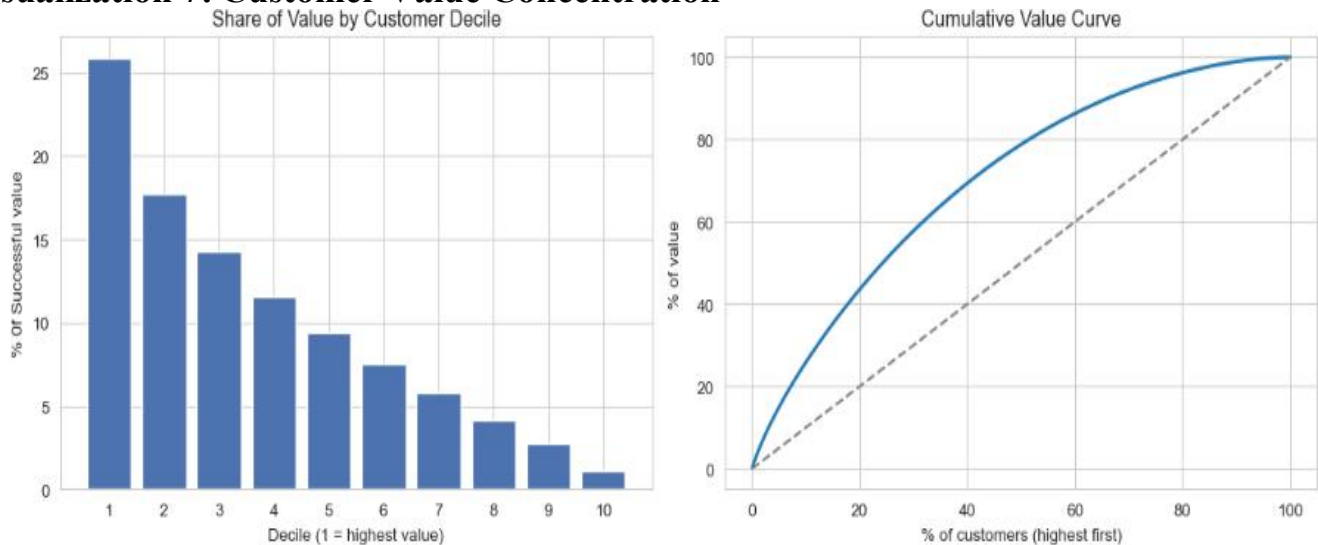


What it shows: Transfers(28.5%) and Cash Withdrawals(25.3%) are flagged for review far more than Bill payments(12.8%).

Why it matters: It shows where review term spend their time most.

Business Insight: Review effort is concentrated on transactions where money actually leaves the account. This is a sensible pattern, not a flaw.

Visualization 7: Customer Value Concentration



{1: 25.86, 2: 17.71, 3: 14.24, 4: 11.57, 5: 9.43, 6: 7.49, 7: 5.77, 8: 4.15, 9: 2.72, 10: 1.06}
Top 10%: 25.9% | Top 30%: 57.8% | Bottom 50%: 21.2%

What it shows: The Top 10% of customers generate 25.9% of all value; the Bottom 50% generate only 21.2%.

Why it matters: Value is unevenly spread across customers.

Business Insight: The top customers decile deserves the most retention attention, since losing them would hurt revenue the most.

Deliverable: [FinTrust_Week2_Data_Analysis.ipynb](#)

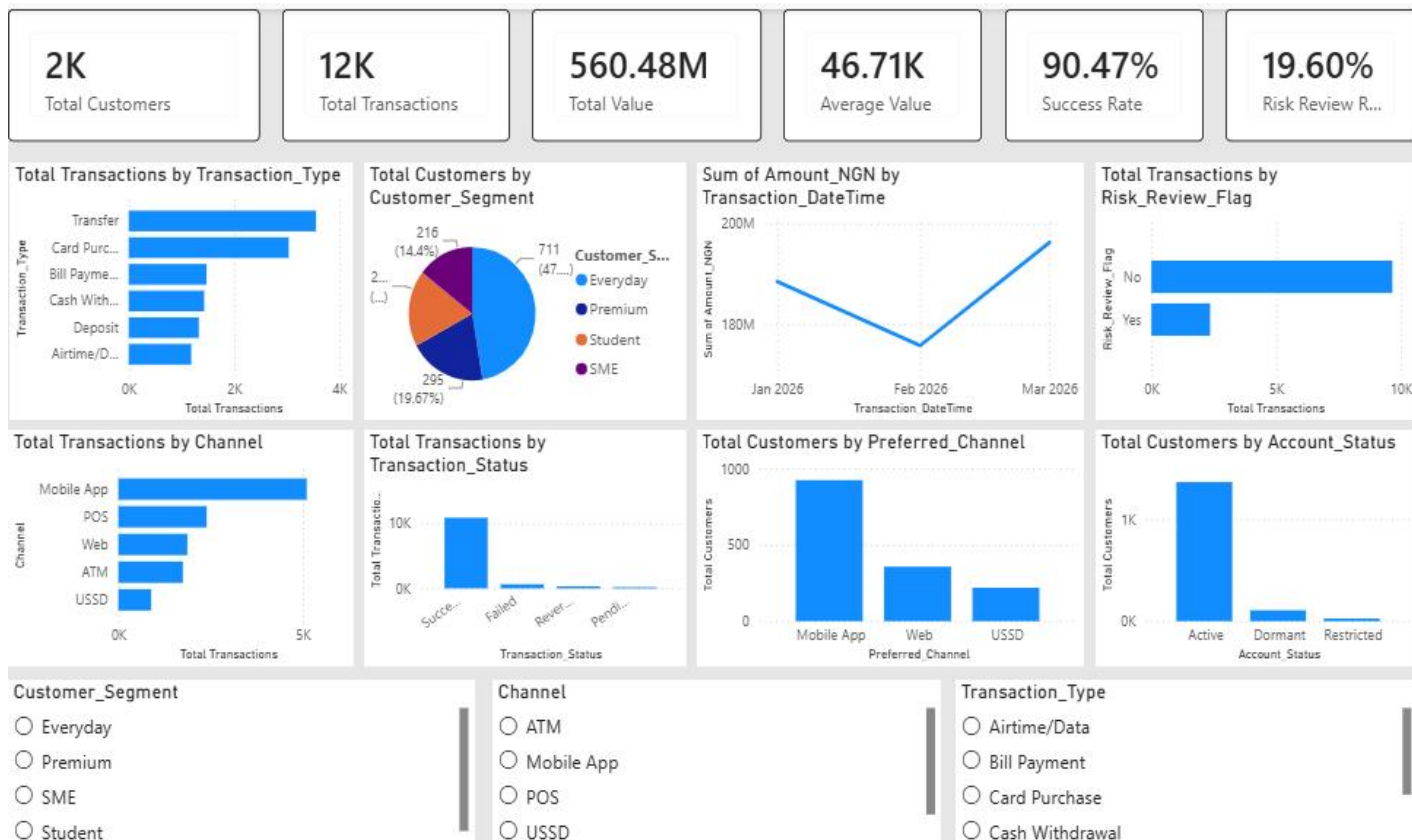
Part D: Power BI Dashboard

An initial management dashboard was developed in Power BI to provide an at-a-glance overview of key performance indicators (KPIs).

Dashboard KPIs:

- Total Customers
- Total Transactions
- Total Transaction Value
- Average Transaction Value
- Transaction Success Rate
- Risk Review Rate

Dashboard Overview



Key Visualizations:

- **Customer Segments:** A donut chart showing the distribution of customers by segment.
- **Transaction Trend:** A line chart showing the total transaction value over month.
- **Transaction Channel:** A bar chart of transaction volume by channel.
- **Risk Review Patterns:** A clustered bar chart showing the risk review by Transaction type.
- **Transaction Types:** A bar chart of total transactions by transaction type.
- **Transaction Status:** A bar chart of total transaction by status.
- **Preferred Channel:** A bar chart of total customers by preferred channel.
- **Account Status:** A bar chart of total customers by account status.
- **Filters/Slicers:** The dashboard includes slicers for Customer_Segment, Channel and Transaction_Type, allowing users to filter all visuals interactively by any combination of these three fields.

Deliverable: [FinTrust Week2 Analytics Dashboard.pbix](#)

Part E: Business Insights

This part seeks to address....

These are the key business insights about the Analysis:

Insight 1. Finding: Restricted and Dormant accounts are still transacting.

➤ **Evidence:** Joining transactions to customer status shows 866 transactions(NGN38.68M) from Dormant accounts and 224 transactions (NGN19.93M) from Restricted accounts.

➤ **Business meaning:** This is a control gap. A “Restricted” account processing normal transactions suggests the restriction isn’t being enforced, which is a fraud/compliance risk worth flagging.

Insight 2. Finding: The “Online” transaction channel has the highest volume but also slightly higher-than-average failed transaction rate.

➤ **Evidence:** Python EDA and Power BI visuals show a high volume of online transactions and noticeable portion of ‘Failed’ statuses.

➤ **Business meaning:** This points to a potential technical or user-experience issue in the online portal that requires immediate investigation to reduce customer friction and lost revenue.

Insight 3. Finding: A small group of customers drives most of the value.

- **Evidence:** The top 10% of customers by spend generated 25.9% of total transaction value, while the bottom 50% generated only 21.2%
- **Business meaning:** FinTrust's top-decile customers deserve the most retention focus, losing a few of them cost more than losing many low-value ones.

Insight 4. Finding: Nearly 1 in 5 transactions gets flagged for risk review.

- **Evidence:** The dashboard shows a 19.60% Risk Review rate (2,352 of 12,000 transactions), against a 90.47% success rate.
- **Business meaning:** That's a heavy compliance workload. Worth checking whether the review rules are broad, since flagging 1 in 5 transactions isn't cheap to sustain.

Insight 5. Finding: A small pocket of customers is maximally engaged.

- **Evidence:** The Customer DQ summary shows 56 of 1,500 customers (3.7%) have a Digital Engagement score of exactly 100.
- **Business meaning:** These are FinTrust's power users, strong candidates for a loyalty program or early access to new features.

Testing & Evaluation:

This section documents how key Week 2 work was checked, following a Work Completed → Test/Evaluation → Finding → Action/Improvement → Result structure.

1. Transaction_DataTime Formatting

Work Completed: Diagnosed and corrected the Transaction_DataTime format issue during data preparation.

Test/Evaluation: Reloaded the cleaned dataset into Power BI and inspected how the date field was recognized.

Finding: The column returned as error field because dates were stored as mm/dd/yyyy while Power BI expected dd/mm/yyyy.

Action Taken: Reformatted the date values before reloading into Power BI.

Result: The field was correctly recognized as a date type, and the monthly transactions trend visual rendered correctly with no data loss.

2. Amount_NGN outlier check

Work Completed: Applied the IQR method of flag potential outliers in Amount_NGN.

Test/Evaluation: Reviewed the 1,474 flagged high_value transactions individually rather than removing them automatically.

Finding: The values reflected genuine high_value transfers and deposits, not data-entry errors.

Action Taken: Retained the records, and used median instead of the mean when reporting a typical transaction size, since the distribution is right-skewed.

Result: The true value distribution was preserved, and the python analysis correctly surfaced the right-skew (Visualization 4) instead of it being masked by removed data.

3. Missing value handling (Device_Type, Location)

Work Completed: Filled missing Devices_Type and Location values in the transaction dataset.

Test/Evaluation: Compared row counts before and after imputation to confirm no records were dropped.

Finding: 96 missing values existed in each field, and row counts matched exactly before and after.

Action Taken: Standardized missing entries to "Unknown" rather than dropping the affected rows or guessing any figure/value.

Result: No transaction records were lost, and "Unknown" remained available as its own category for channel/device analysis instead of introducing bias.

Limitations & Assumptions

- All FinTrust customer, transaction and Risk_Review_Flag data is synthetic. Findings describe patterns in the dataset only; the Risk_Review_Flag is not a real fraud or credit_risk determination.
- The dataset covers only three months (Jan-Mar 2026). This is too short window to confirm seasonality or long-term growth, so the month-to-month comparison in SQL question 5 should be read as a data limitation rather than a business trend.
- Transaction timestamps are evenly spaced within each month (a constant ~ 133 transactions/day). This is a known characteristic of the synthetic data generation rather than real customer behavior.
- The existing Customer_Segment labels do not explain differences in transaction to revisit; a behavior-based segmentation may be more useful in later weeks.
- Restricted and Dormant accounts still show transaction activity (866 transactions from Dormant accounts, 224 from Restricted accounts). This was identified as a data/control observation during analysis.
- Data cleaning, SQL, Python, and Power BI outputs were each validated independently per tool rather than through a single automated pipeline, so consistency across outputs was checked manually.